

Daily Geometry Drill #7

Sheet by Henry Koduthore

Section	Topic	Questions	Target time
A	Rapid Recognition	10	2 min 30 sec
B	Manipulation Drills	10	8 min
C	Substitution & Structure	8	10 min
D	Challenge	5	15 min

New toolkit today: Loci of points · Apollonius circle $\frac{PA}{PB} = k$ · geometry-logic conditionals · flaw-spotting in synthetic proofs.

Attempt each question before checking answers. Time yourself. No calculators.

Section A Rapid Recognition

(≤15 s each)

Name the locus straight off; only the standard families occupy these questions.

- Write down the locus of points whose distance from the origin is 5.
- Write down the locus of points equidistant from the two points (0,0) and (4,0).
- The locus of P with $PA = 2PB$, where $A = (0,0)$ and $B = (6,0)$, is a circle. Write down the x -coordinate of its (relevant) endpoint on the x -axis where $y = 0$ between A and B .
- Write down the locus of points that are always 3 units from the fixed line $y = 4$.
- The locus of points whose distance to the fixed point (0,1) equals their distance to the fixed line $y = -1$ is a parabola. Write down a point on it other than the origin with integer coordinates.
- The locus of points whose distance to the point (1,2) is exactly 3 is a circle. Write down its equation.
- Write down the locus of the midpoint M of every radius OQ of the circle $x^2 + y^2 = 4$ as Q runs around the circle (with O the origin).
- The equation $x^2 + y^2 = 2x$ describes a circle. Write down its radius.
- Write down whether the statement “ $x^2 + y^2 = 4$ implies $x = \pm 2$ or $y = \pm 2$ ” is true or false.
- Write down the locus of points P with $PA = PB$, where $A = (1,3)$ and $B = (5,3)$. (Give its equation in x .)

Section B Manipulation Drills

(≤50 s each)

Square both sides to force Apollonius loci into circle form, or reason about necessity/sufficiency for the logic items.

- The circle of Apollonius for $PA = 2PB$ with $A = (0,0)$, $B = (6,0)$ has radius?
A) 4

- B) 6
C) 8
D) 12
2. Locus of points equidistant from $(0, 0)$ and $(0, 4)$ is?
A) $x = 2$
B) $y = 2$
C) $y = 4$
D) $x = 4$
3. The points satisfying $x^2 + y^2 \leq 4$ form a filled disk. Its area is?
A) 2π
B) 4π
C) 8π
D) 16
4. The distance from $P(x, y)$ to the origin is $\sqrt{x^2 + y^2}$. The locus $\sqrt{x^2 + y^2} = 1$ is a circle. Write down its circumference.
5. A point is known to lie strictly inside the unit disk $x^2 + y^2 < 1$. Which of the following statements is *necessarily* true for such a point?
A) $x^2 + y^2 < \frac{1}{2}$
B) $x + y < 1$
C) $x^2 + y^2 < 2$
D) $x^2 + y^2 > 0$
6. The Apollonius locus $PA = 3PB$ with $A = (0, 0)$, $B = (2, 0)$: what is the x -coordinate of its right endpoint on the x -axis?
A) 3
B) 4
C) 6
D) 9
7. To prove a point P lies on the unit circle $x^2 + y^2 = 1$, which single condition is *sufficient* on its own?
A) $x^2 + y^2 < 1$
B) $x = -\frac{1}{2}$ and $y = \frac{\sqrt{3}}{2}$
C) P is not the origin

D) $x + y = 1$

8. The locus of P with $PA = PB$ where $A = (0, 0)$ and $B = (6, 0)$ is the perpendicular bisector $x = 3$. Write down the distance from $(3, 0)$ to $(0, 0)$.
9. A circle has equation $x^2 + y^2 - 4x - 6y + 9 = 0$. Write down its centre.
10. Write down the radius of the circle $x^2 + y^2 - 4x + 6y + 9 = 0$.

Section C Substitution & Structure

(≤ 75 s each)

Assemble loci, equations and logic: square, complete the square, and decide whether each borrowed fact is necessary, sufficient, or both.

1. The locus of P with $PA = PB$, where $A = (0, 0)$ and $B = (6, 0)$, is the line $x = 3$. What is its distance from B ?
- A) 3
B) 6
C) 9
D) 0
2. The circle of Apollonius $PA = 2PB$ with $A = (0, 0)$, $B = (6, 0)$ has centre?
- A) $(8, 0)$
B) $(4, 0)$
C) $(2, 0)$
D) $(-8, 0)$
3. The inequality $x^2 + y^2 \leq 9$ describes a filled disk. Which point is strictly inside it?
- A) $(2, 2)$
B) $(3, 3)$
C) $(2, 3)$
D) $(4, 0)$
4. The locus $\frac{PA}{PB} = 1$ (equal distances) with $A = (0, 0)$, $B = (6, 0)$ is a line. What is it?
- A) $x = 3$
B) $x = 6$
C) parallel to the y -axis through B
D) $y = x$
5. Given the circle $x^2 + y^2 = 5$, how many of the following points lie on it: $(1, 2)$, $(2, 1)$, $(\sqrt{5}, 0)$, $(1, -2)$?
- A) 3

- B) 4
C) 2
D) 1
6. The locus of P with $PA + PB = 8$ and $A = (-3, 0)$, $B = (3, 0)$ is an ellipse. The length of its major axis is?
A) 4
B) 8
C) 6
D) 16
7. A point is said to be “well-placed” if it lies on the unit circle. Which statement is *both necessary and sufficient* for the point (a, b) to be well-placed?
A) $a^2 + b^2 = 1$
B) $a + b = 1$
C) $a^2 + b^2 < 1$
D) the point is not the origin
8. The circle $x^2 + y^2 - 4x + 4 = 0$ has radius? (Complete the square.)
A) 1
B) 2
C) 0
D) 4

Section D Challenge Ramp(≤ 90 s each)

Stack the whole week: a locus, a completing-the-square circle, and a clean TRUE/FALSE verdict on a geometric claim.

1. The Apollonius locus $PA = 2PB$ with $A = (0, 0)$, $B = (6, 0)$ has centre $(8, 0)$ and radius 4. Find the locus of points P with $PA = KB$ obtained when the ratio is 2: the circle's equation is $(x - 8)^2 + y^2 = 16$. What is the distance between the two x -axis endpoints of this locus?
A) 8
B) 4
C) 16
D) 6
2. The circle with equation $x^2 + y^2 - 4x - 6y + 12 = 0$ has radius?
A) 1
B) 2
C) 3

D) $\sqrt{12}$

3. Which of the following is *sufficient* for the point (x, y) to lie *strictly inside* the unit disk $x^2 + y^2 < 1$?

A) $x^2 + y^2 < \frac{1}{2}$

B) $x < \frac{1}{2}$ and $y < \frac{1}{2}$

C) $x^2 + y^2 < 2$

D) $x > 1$ and $y > 1$

4. Fil finds the circle of Apollonius $PA = 3PB$ with $A = (0, 0)$ and $B = (2, 0)$ by squaring $\frac{PA}{PB} = 3$ and claiming the locus is $(x - 2)^2 + y^2 = 9$. Where is the error in his conclusion?

A) His equation is a circle centred at B with radius 3, but the true locus must involve the distance to A as well; expanding $x^2 + y^2 = 9((x - 2)^2 + y^2)$ gives a different circle.

B) Squaring loses the sign information, so the locus is really a single point.

C) The set $PA = 3PB$ is a straight line, not a circle at all.

D) There is no error; $(x - 2)^2 + y^2 = 9$ is correct.

5. The circle passes through the point $(1, 2)$ and has centre $(0, 0)$. Write down its equation.

“A locus is a sentence the points all agree on; a geometric proof is only as strong as the condition you actually used.”

Found a mistake or want to contribute a sheet?

Visit the open-source project at github.com/The-CerealDev/speed-maths